

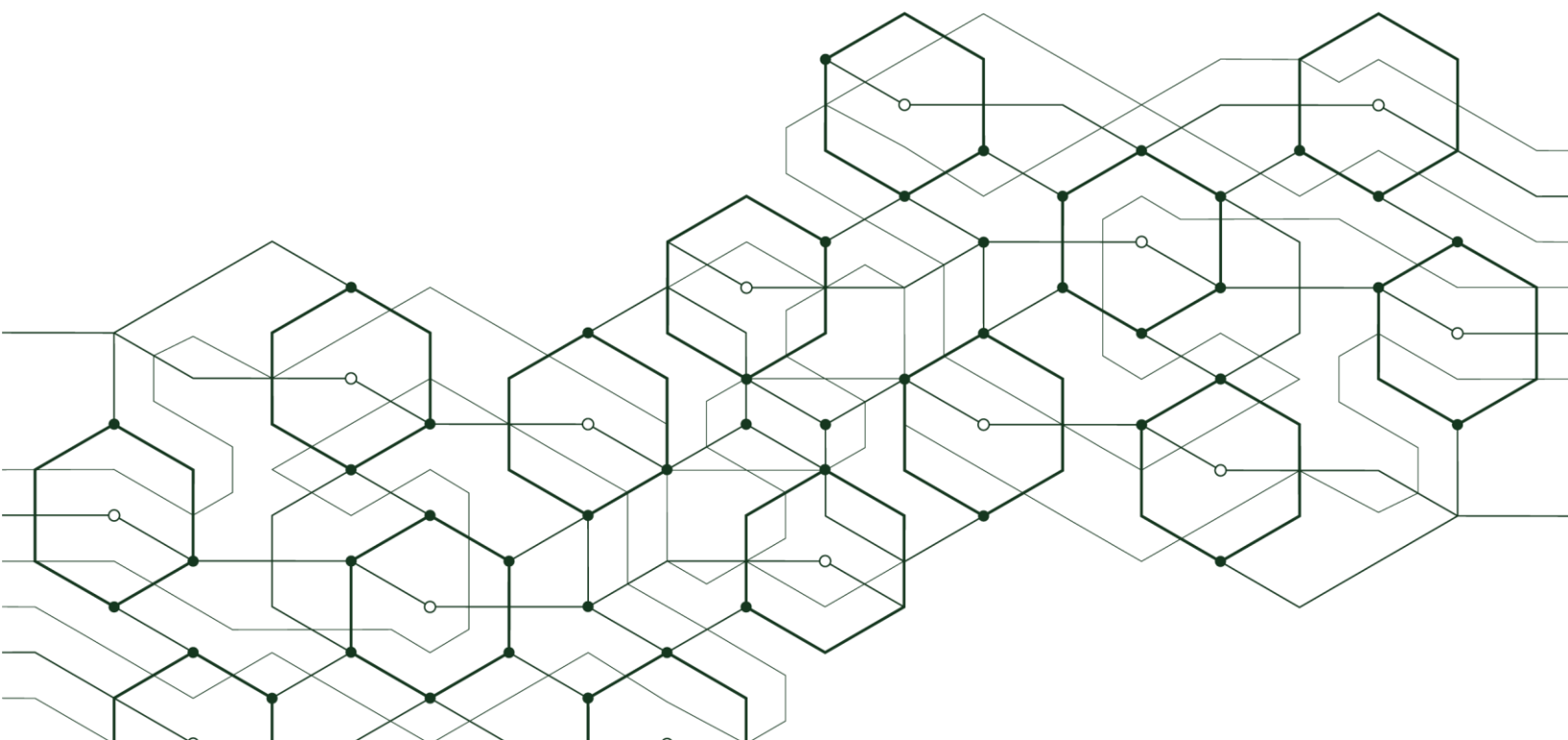
Java in the Era of AI & Cloud-Native Innovation

Empowering Developer Productivity & Enterprise Agility

by **Chris Rommel**, *EVP*

License to Distribute:

ORACLE®



Introduction

Enterprise organizations must adapt. New corporate initiatives and application workloads are redefining the metrics and means of success. Organizations are prioritizing cloud-native modernization and AI integration to support scalability, responsiveness, and innovation. As platforms for both development as well as end-user engagement, the cloud and AI are presenting organizations with unmatched opportunities for growth and redefinition – but those organizations must have the tools and technologies in place needed to successfully adapt.

Development timelines and cost pressures are forcing organizations to reevaluate their incumbent technologies and identify trusted partners capable of providing reliable and adaptive platforms for innovation. While AI and cloud-native development are transforming development practices and organizational demands, software languages remain the foundation of organizational transformation and future adaptability.

With over three decades of ecosystem progress and investments, Java has remained the leading choice for enterprise software development. Java remains critical across industries, including finance, industrial, defense, and telecommunications, with widespread use in both greenfield and legacy environments. Java, guided by Oracle's stewardship, is proving essential in meeting new development challenges, boosting productivity, and enabling scalable enterprise-grade solutions.

Java Leading Language Choice:

- The majority of enterprises rely on Java for their development needs
- Java is the #1 language for cloud-native development
- Java is #1 for AI developer productivity
- 77% of Java developers meet or beat deadlines

Background on This Primary Research

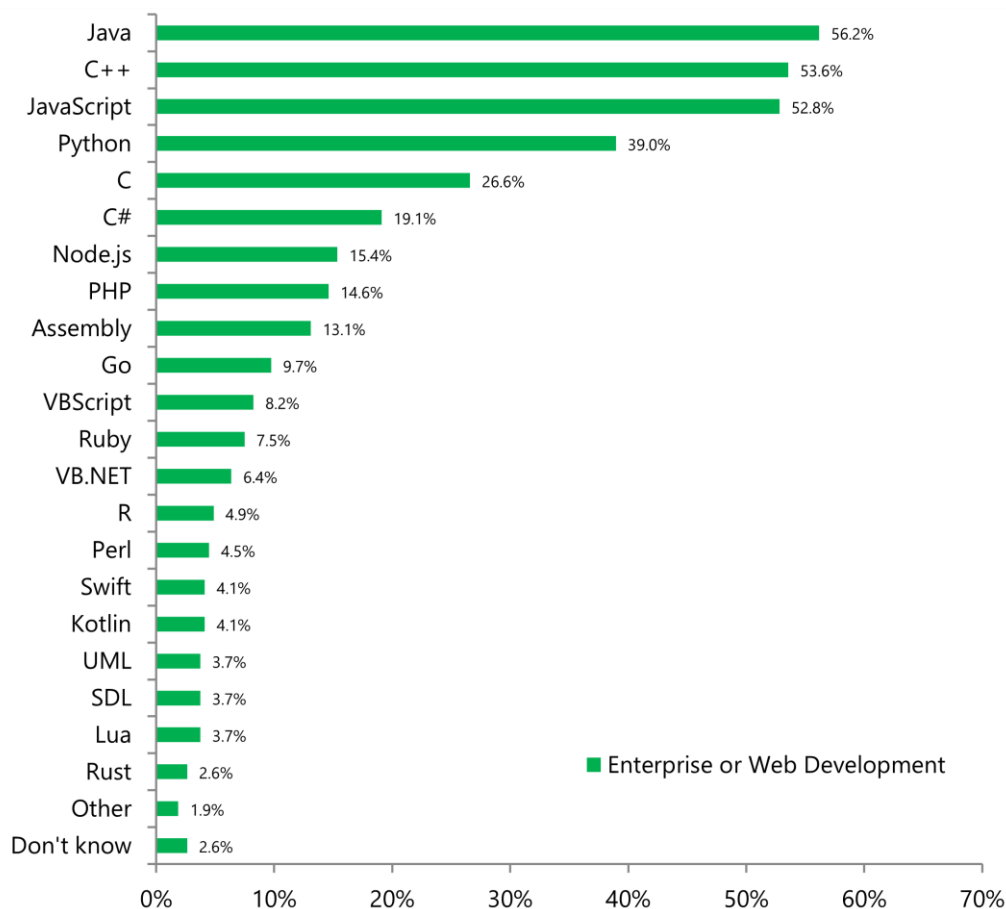
VDC Research has been covering the software development technology market since 1994. The analysis and supporting discussions in this paper are based on VDC's ongoing research in this market and findings from a 2025 survey of 532 technology executives, managers, and developers. This global survey offers insight into leading business and technical trends impacting development organizations as well as the best practices and technologies implemented to address them.

Java: The Leading Choice for Enterprise Innovation

Against a landscape of perpetual change in business climates and technical goals, Java and the ecosystem around it have emerged as a key ingredient for successful enterprise software development. The nature of highly-specialized application requirements creates value and use cases for a number of languages for software development. For example, the use of other object-oriented and scripting languages such as C++ and Python has gained additional traction within the IoT and mobile segments, displacing legacy choices such as C and Assembly. However, Java's incumbent market leadership has only increased among professional developer ranks. In fact, 56% of enterprise developers reported Java's use within their organization, an increase from 53% in 2023. Java's consistent and increasing use in recent years speaks to its utility and entrenchment within development organizations worldwide.

Exhibit 1: Programming Language Use within Enterprise/IT Organization

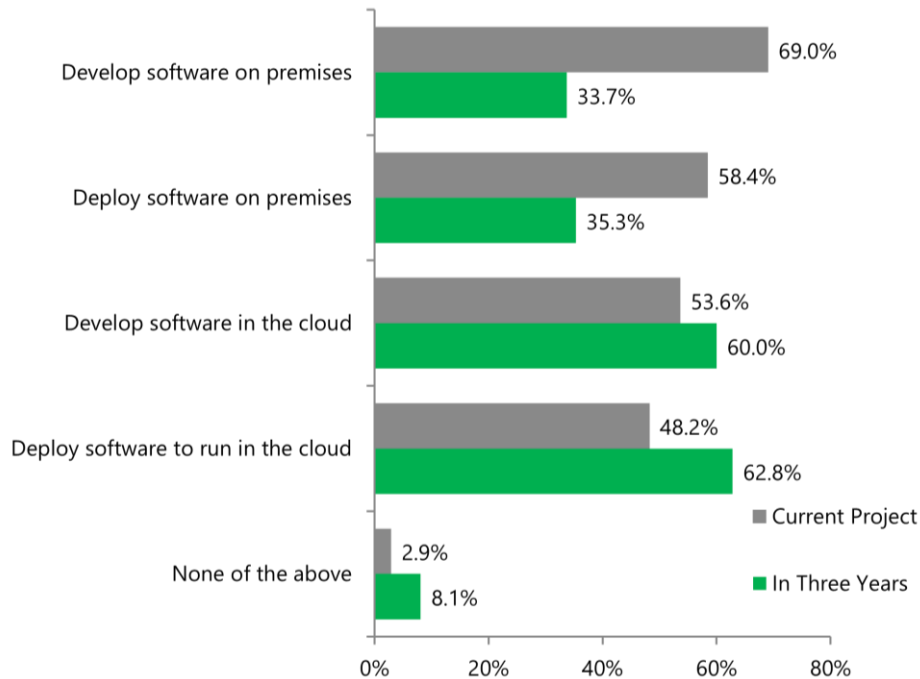
(Percentage of Respondents)



Shift to Cloud-Native Java:

Exhibit 2: Cloud Use: Current Use & Future Projects

(Percentage of Respondents)



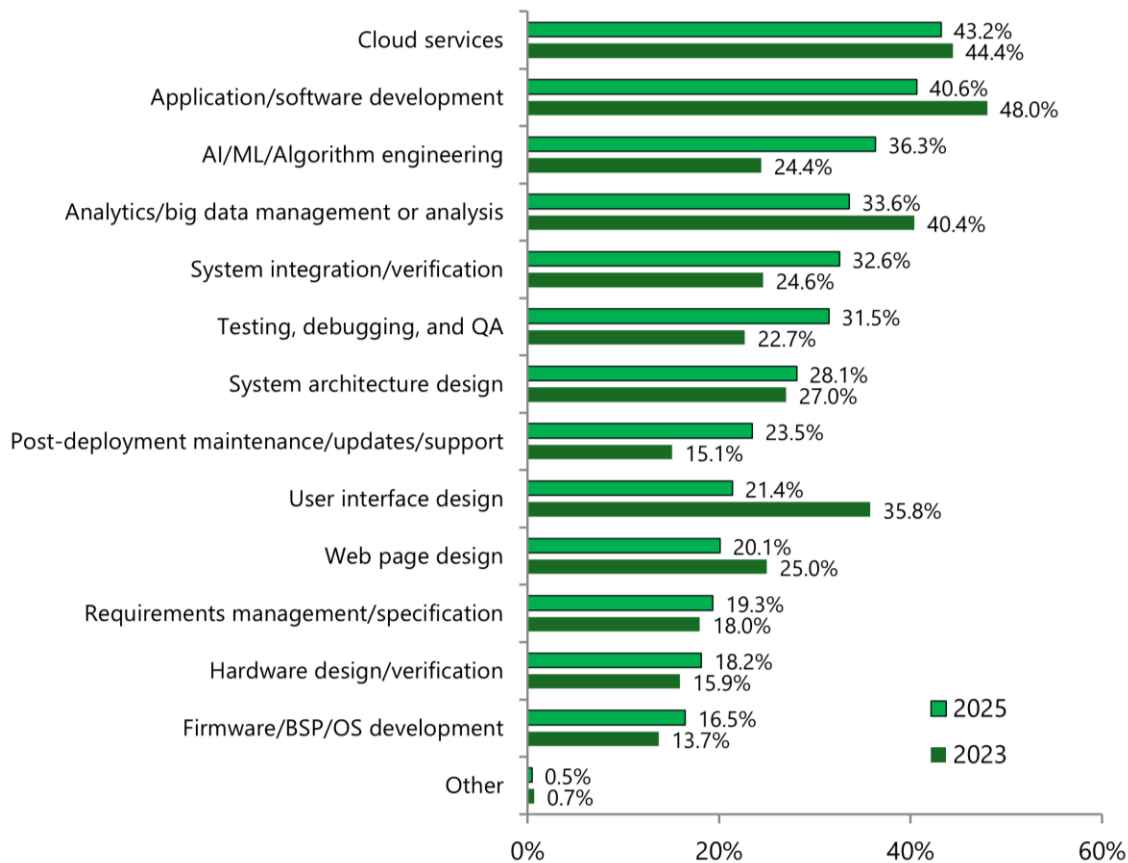
While some may consider cloud-based development and infrastructure a fundamental component of today's enterprise best practice, adoption and progress are still uneven. Many organizations, influenced by backward-looking project requirements and overly conservative management, are still early in their adoption journey. Other early adopters are still working to find the best recipe of processes, technology, and partners to help them scale efficiently to meet the next generation of their productivity and functionality goals.

Cloud services emerged as the most frequent area of engineering investment

While adoption remains uneven, cloud use will saturate development practices over the next three years. While nearly 70% of respondents still develop on premises (at least in part), only a third expect to do so in three years. Meanwhile, the majority also expect to be deploying their software to run in the cloud in that timeframe. In addition, cloud services emerged as a key point of additional internal focus for development organizations. In fact, cloud services surpassed application software development as the top task with planned increase in internal investments in the 2025 study [See Exhibit 3].

Exhibit 3: Areas of Engineering Task Investment: Current Use & Future Projects

(Percentage of Respondents)



Future-Proofing Through Platform Maturity

Oracle's continued investment in the cloud ecosystem can be seen both via its product portfolio, as well as its quantifiable presence within the ecosystem at large. For example, Project Leyden and virtual threads enable lower overhead and higher performance for container-based Java apps. In addition, the company's investment in and Java's integration with its Oracle Cloud Infrastructure affords development organizations the ability to easily integrate with existing multi-cloud deployments as well as to design and run cloud-native applications and third-party, open source content. VDC Research projects that the number of deployed Java Virtual Machines (JVMs) will surpass 100 billion by 2029, with over 80% running in cloud environments. This growth underscores the trust developers hold in the security, reliability, and robust support of Oracle's Java ecosystem.

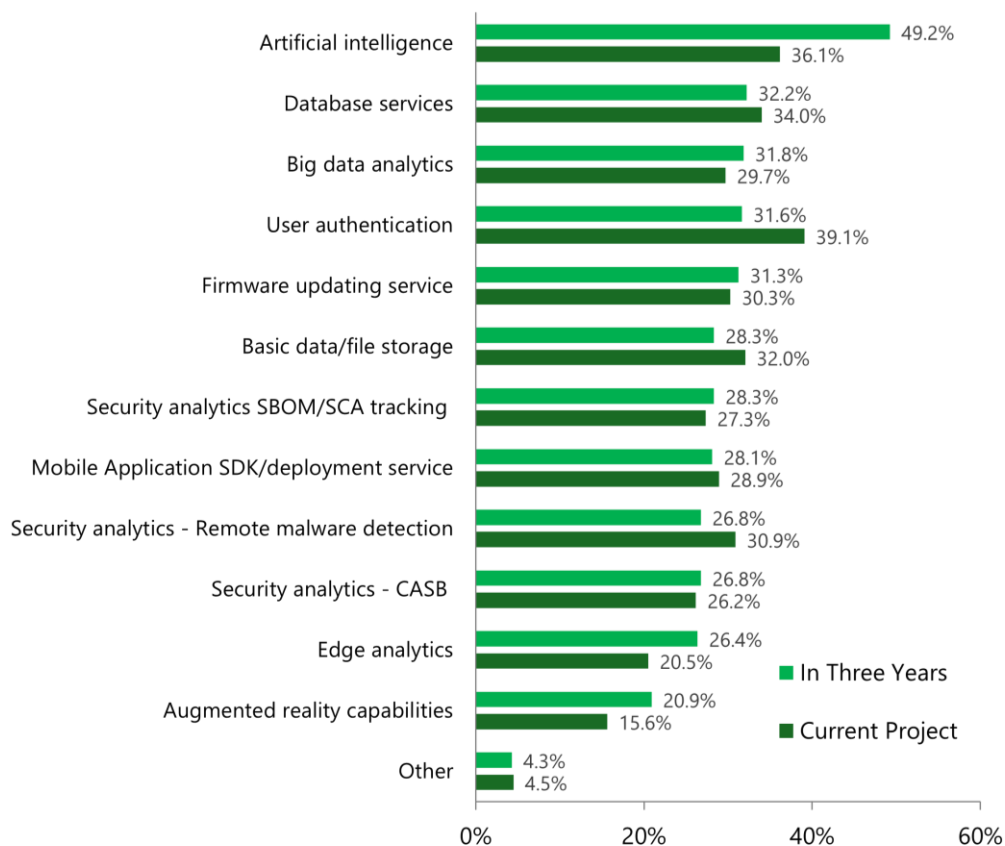
Oracle's continued investment in the cloud ecosystem has yielded constant improvements in both the community's efficiency as well as its recognition of Java's value:

- Java was the most-frequently used language by developers building applications on the cloud.
- 84% of developers using Java stated that the language helped their organization efficiently develop cloud-native applications – an increase from 79% in 2023
- In addition, 75% agreed that Java helps speed cloud development – an increase from 69% in 2023
- 84% stated that the Java ecosystem offers resources to help cloud application development

The cloud's impact and influence on the future of software development trends and investments will only grow going forward. Moreover, its position as a foundation and fuel for future software innovations will underscore the importance of choosing partners that can help deliver the needed functionality to ensure dependency and efficiency. For example, while we noted that cloud services were the engineering task with the highest rate of planned future investment, AI/ML was the area which saw the highest increase in the past two years (36% in 2025 vs. 24% in 2023), highlighting the rapid increase in organizational focus on that domain. Furthermore, when developers were asked what cloud services they expected to run in three years, Artificial Intelligence was cited as the overwhelming top choice. For resource intensive workloads, the scalable infrastructure and faster deployment make cloud and AI a natural pairing [See Exhibit 4].

Exhibit 4: Cloud Services: Current Use & Future Projects

(Percentage of Respondents)



AI and Developer Productivity

The use of AI is revolutionizing software development. AI development workloads are now a top-five initiative among enterprises surveyed for this research. Not only are AI workloads redefining software stack and hardware system requirements, but generative AI is also transforming the means and expectations for software development. AI code assistants and IDE integrations can improve both code quality and delivery speed. Already, more than 50% of developers use GenAI services for code generation with adoption expected to increase going forward. However, experimentation alone does not always translate into greater developer efficiency or improved code quality. Selecting the right technology partner to support these efforts and ensure alignment with this critical industry trend is essential.

Java is the #1 language for AI Developer Productivity

In recent years, Java emerged as a trusted pillar supporting the foundation of enterprise AI adoption. Its long-standing ubiquity, strong typing, and reliability – combined with Oracle’s commitment to readability, versatility, and compatibility – place it squarely at the forefront of this next wave of enterprise investment. Already, developers using Java are more sophisticated and advanced in their pursuit of AI business goals. For example, developers using Java report lower reliance on hand-coded assets as well as a higher rate of use of GenAI and pretrained AI models, outpacing developers using other languages for their software development. Moreover, when surveyed developers were asked to identify which language used was the best for AI developer productivity (e.g., code assistance, business logic service), Java was selected more frequently than any other option. Likewise, 64% of Java users leverage pretrained AI models, outpacing the use rate of other developers. With use cases for GenAI for tasks from design through test, solutions like Oracle Java Management Service (JMS) enhance visibility into code quality, performance regressions, and license optimization.

The Reliable Business Case for Java

While technology trends and organizations' pursuit of innovation often drive investigation of new development solutions, adoption decisions invariably rest on return on investment. In both this study as well as in VDC's 2023 market analysis, developers using Java report multiple advantages from using the platform. In fact, 77% of Java developers meet or beat deadlines, compared to 67% of non-Java users, a critical advantage for time-sensitive projects. Moreover, projects not using Java are twice as likely to face delays of 3 to 6 months or more (14%) compared to those using Java (7%). Java offers development teams opportunities for cost savings and accelerated product delivery through platform consistency, skilled labor availability, and widespread compatibility across cloud and edge infrastructure.

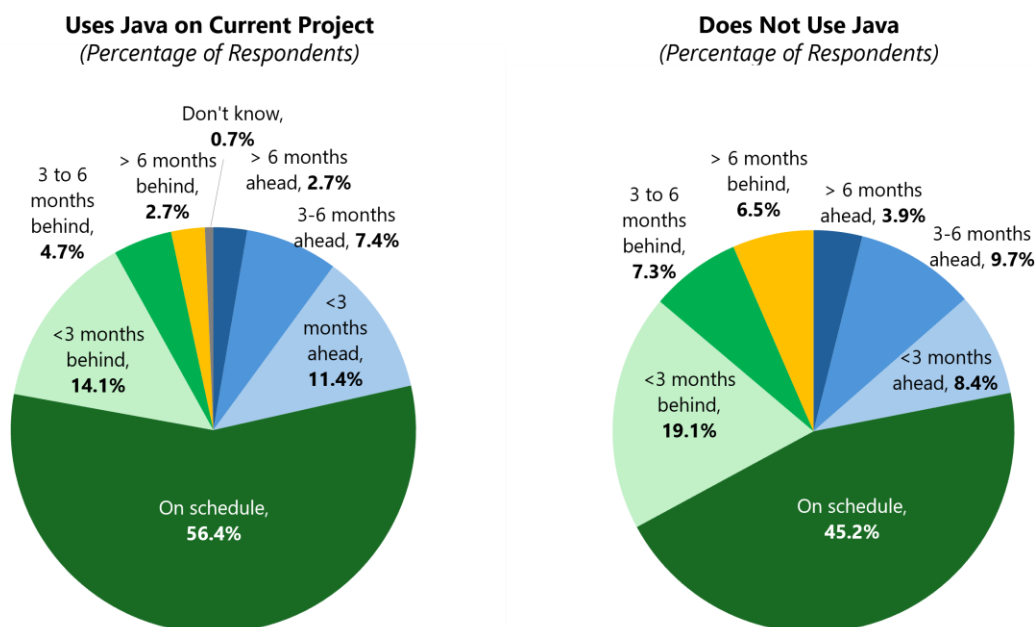
Reduced Risk and Better Project Outcomes

Java presents enterprises with a range of solutions to help accelerate development and maximize the capabilities of their development projects. For example, Java's mature libraries and tools reduce in-house effort, enable faster onboarding, and leverage the best innovations that its open ecosystem has to offer. Java enables better project outcomes, especially in regulated, high-security environments (e.g., government, financial services).

77% of Java developers meet or beat deadlines

Project Schedule Adherence

(Percentage of Respondents)



Java's Cross-Sector Strength

Java presents development organizations a powerful platform for addressing the top tech trends. In our 2025 survey, over 500 developers and technology leaders rated the languages used by their organization in their utility versus the leading tech trends being pursued by their organization. Java was in the top three most-important cited languages for 14 out of the 15 technology trends evaluated. Notably, Java achieved #1 rankings for cloud-native development, microservices, AI developer productivity, and Big Data, showcasing its utility for next-generation, data-intensive development initiatives.

In addition, engineers cite high satisfaction with Oracle Java in terms of reliability, tooling, and support—reducing risk and churn during AI development and cloud migrations. Oracle's roadmap transparency and support services ensure engineers can plan for long-term innovation.

VDC's Summary & Recommendations

Future Innovation Driven by Present Software Choices

The accelerating demand for cloud-native architectures and AI-enhanced development requires mature, secure, and scalable platforms—making Java more essential than ever. Java is evolving to meet the demands of cloud-native, AI-driven enterprises. Not only is Java the most widely-used language for enterprise software development, but it is also the top-rated solution for both cloud-native development as well as AI developer productivity.

With robust tools, trusted support, and a roadmap aligned with developer needs, Java – particularly Oracle Java – is uniquely positioned to enable next-generation applications. In order to best position themselves for success, enterprises should not only invest in Java-centric toolchains for cloud-native and AI development but also look towards the investments Oracle has made directly to support these next-generation trends. For example, Oracle's OCI-based deployment infrastructure can provide a sound foundation for AI-focused apps that can leverage ONNX models or Deeplearning4j. Ultimately, enterprises have recognized Java as a key gear to unify cloud, AI, and enterprise workload development.

While niche technologies may offer advantages for specific use cases, today's dynamic business environment demands robust, versatile solutions that can support a broad spectrum of mission-critical applications. Java remains the trusted foundation for enterprises and the go-to language for developers building the next generation of innovative solutions.

About the Author

Chris leads VDC's research programs and consulting engagements focused on development and deployment solutions for intelligent systems. He has helped a wide variety of clients respond to and capitalize on the leading trends impacting next-generation industrial and device markets, such as security, the IoT, and engineering lifecycle management solutions. Chris holds a B.A. in Business Economics and a B.A. in Public and Private Sector Organization from Brown University.

Email Chris at crommel@vdcresearch.com

About VDC Research

Founded in 1971, VDC Research provides in-depth insights to technology vendors, end users, and investors across the globe. As a market research and consulting firm, VDC's coverage of development and engineering technologies helps our clients make critical decisions with confidence. Offering syndicated reports and custom consultation, our methodologies consistently provide accurate forecasts and unmatched thought leadership for deeply technical markets. Located in Southborough, Massachusetts, VDC prides itself on its close personal relationships with clients, delivering an attention to detail and a unique perspective that is second to none.

For more information, contact us at info@vdcresearch.com.